Focused Adapter: Enhancing Fine-Grained Attention for Remote Sensing Image–Text Retrieval

Wen-Liang Du, *Member, IEEE*, Xiao-Yu Xu, Jiaqi Zhao, *Member, IEEE*, Rui Yao, *Member, IEEE*, Xiaoyi Li, and Yong Zhou

***Abstract*—The emergence of large-scale vision–language models (VLMs) has significantly advanced remote sensing image–text retrieval (RSITR) by providing powerful cross-modal semantic priors. However, when adapted to the remote sensing (RS) domain, these models struggle to capture fine-grained representations due to their inherent global semantic bias. To address these limitations, we propose the focused adapter (FocA), a plug-and-play parameter-efficient fine-tuning (PEFT) architecture designed to enhance fine-grained perception from frozen VLMs. The FocA features a hybrid structure consisting of two components: an enhancement adapter and an alignment adapter. The enhancement adapter utilizes bottleneck self-attention to capture rich patch-level details often lost during global pooling. The alignment adapter incorporates a cross-modal shared projection subspace to facilitate early feature interaction and implicit alignment. In addition, we develop an explicit shared loss to provide direct semantic supervision, preventing the dilution of critical features during deep propagation. Extensive experiments on the RSICD and RSITMD datasets demonstrate that FocA achieves state-of-the-art (SOTA) performance, attaining a mean recall (mR) of 37.61% on RSICD and 48.95% on RSITMD. Our code is available at https://github.com/WenliangDu/FocA**

***Index Terms*—Fine-grained attention, parameter-efficient fine-tuning (PEFT), remote sensing image–text retrieval (RSITR), vision–language models (VLMs).**

I. Introduction

THE availability of earth observation data has increased substantially as remote sensing (RS) technology continues to advance [1], [2]. This surge has catalyzed significant research into RS image–text retrieval (RSITR), a cross-modal task that bridges the gap between visual and textual modalities, enabling applications such as disaster response, urban planning, and environmental monitoring.

Traditional research on RSITR mainly focuses on two main directions: global feature alignment and global-local alignment. Global feature alignment methods project images and sentences into a shared embedding space, minimizing the distance between corresponding pairs. Abdullah et al. [3] proposed a deep bidirectional triplet network utilizing ResNet [4] and LSTM [5] to project visual and textual features into a unified Hamming space. Ji et al. [6] introduced a knowledge-aided momentum contrastive learning (KAMCL) framework to refine global feature boundaries by maintaining a dynamic queue of negative samples. However, global feature alignment methods treat images as coarse-grained representations, overlooking local details and failing to capture fine-grained semantics. To address this limitation, global-local alignment methods have been developed that integrate regional features with global representations for finer-grained alignment. Yuan et al. [7] developed a global-local information fusion method that integrates local region representations with global features through dynamic fusion to capture both coarse-grained context and fine-grained details. Ma et al. [8] proposed an *a*-direction-oriented visual-semantic embedding (DOVE) model that utilizes regional visual features to direct the alignment of visual and textual representations in the latent space. Nonetheless, the feature fusion process often compromises local information integrity and incurs substantial parametric overhead. Moreover, constrained by limited model capacity, traditional methods often struggle to capture rich semantic representations [9], [10].

Pretrained models such as contrastive language image pretraining (CLIP) [11] offer powerful, generalizable visual–language representations learned from vast amounts of natural image–text pairs, providing a strong foundation for cross-modal retrieval tasks and enabling more effective domain adaptation when properly fine-tuned. Consequently, numerous recent studies have focused on adapting powerful vision–language models (VLMs) [11], [12] to the RS domain. Zhang et al. [13] introduced GeoRSCLIP by fine-tuning the pretrained CLIP model on the RS5M dataset, narrowing the domain gap between natural and RS images. To improve efficiency, parameter-efficient fine-tuning (PEFT) methods, such as adapters [14], [15], and prefix-tuning [16], have emerged. Huang [17] proposed HarMA, which employs a gating-based adapter mechanism to adaptively balance

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Wen-Liang Du, Xiao-Yu Xu, Jiaqi Zhao, Rui Yao, and Yong Zhou are with the School of Computer Science and Technology, School of Artificial Intelligence, the Mine Digitization Engineering Research Center of the Ministry of Education, and Jiangsu Provincial Industrial Technology Engineering Center for Intelligent Sensing and Emergency IoT in Underground Space, China University of Mining and Technology, Xuzhou 221116, China (e-mail: wldu@cumt.edu.cn; xiaoyuxu@cumt.edu.cn; jiaqizhao@cumt.edu.cn; ruiyao@cumt.edu.cn; yzhou@cumt.edu.cn).

Xiaoyi Li is with the RIOH High Science and Technology Group, Beijing 100088, China (e-mail: lixiaoyi@hstg.com.cn).

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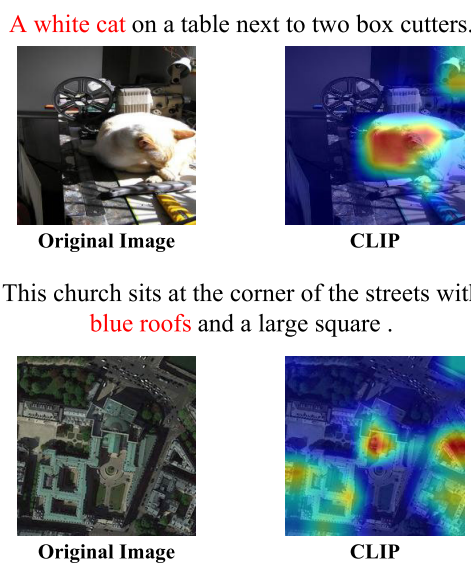


Fig. 1. Visual evidence of the limitation in fine-grained semantic alignment when adapting CLIP to RS imagery using Grad-CAM. (Top) In object-centric natural images (query "a white cat"), CLIP establishes precise visual-semantic alignment, focusing exclusively on the prominent subject. (Bottom) In scene-centric RS images (query "blue roofs"), the model fails to distinguish the specific color attribute due to the interference of dense objects, incorrectly activating adjacent gray roofs. This highlights that CLIP's global aggregation mechanism is inadequate for capturing fine-grained entities in complex RS scenes.

preservation of pretrained knowledge with the integration of domain-specific features using minimal trainable parameters.

Nevertheless, directly adapting CLIP models to RSITR faces a fundamental challenge in fine-grained semantic alignment. This challenge arises because CLIP is pretrained on object-centric natural images taken at close range from a frontal perspective; these images usually feature sparse and large objects that are easy to distinguish based on fine-grained semantics. On the contrary, RS images are scene-centric and captured from a distant, top-down angle. They often contain dense small objects, texture-dominated regions, and large intraclass variation, which make fine-grained semantic discrimination more difficult [18]. In general, fine-grained semantics refers to two complementary aspects: attribute-level and relational-level details. Attribute-level details describe individual object properties, such as color ("blue roofs"), object size ("large lawns"), and numerical features ("three lines of cars"). Relational-level details refer to the spatial and topological relationships between objects, including spatial arrangement ("next to buildings") and relative positioning.

Fig. 1 illustrates an example of fine-grained semantic alignment when adapting CLIP to both natural and RS imagery. The right column in the figure shows gradient-weighted class activation mapping (Grad-CAM) [19] visualizations of CLIP for the highlighted red fine-grained text. CLIP aligns prominent, object-centric subjects effectively in natural images. However, it struggles with RS images that contain multiple dispersed objects and dense fine-grained features. In particular, when queried with "blue roofs," the model may incorrectly activate adjacent gray roofs due to a lack of precise fine-grained semantic alignment. From the perspective of Transformer architectures, visual details and geometric shapes are best preserved in shallow layers [20]. However, increasing network depth leads to the global aggregation mechanism inherently "diluting" and "smoothing" local details into unified global tokens, resulting in semantic oversmoothing [21], [22].

To bridge this granularity gap, we propose the focused adapter (FocA), a plug-and-play PEFT architecture designed to enhance fine-grained attention capabilities in frozen models. In this adapter-based PEFT paradigm, adapters transform the intermediate hidden states along the hierarchical pathway to preserve hierarchical representations at different Transformer depths [17], [23], [24], thereby effectively capturing texture-dominated content and dense small objects inherent to RS imagery for lightweight cross-modal alignment. To further advance the alignment of cross-modal fine-grained representations, FocA proposes enhancement and alignment adapters, each specifically tailored to address both attribute-level and relational-level details. The enhancement adapter leverages intramodal bottleneck self-attention to enhance the representation of attribute-level details that are usually lost during global pooling. This self-attention also models the spatial contexts among feature patches, effectively capturing relational-level topological structures. Subsequently, the Alignment Adapter constructs a cross-modal shared projection to enable feature alignment. This mechanism ensures that both attributes and relationships are implicitly aligned across multiple scales. Finally, we introduce an explicit shared loss function to enforce semantic constraints on intermediate layers to further preserve fine-grained alignments during the global aggregation process.

In summary, our main contributions are as follows.

1) We propose FocA, a plug-and-play fine-tuning architecture with a distinctive hybrid adapter structure that effectively addresses the fine-grained alignment challenge in RSITR while maintaining high parameter efficiency.
2) We introduce an explicit shared loss to enforce semantic constraints on intermediate representations. This strategy mitigates the semantic oversmoothing in deep Transformers by aligning modalities where fine-grained details are most salient, preventing dilution during global aggregation.
3) Extensive experiments on benchmark datasets, including RSICD and RSITMD, demonstrate that FocA achieves state-of-the-art (SOTA) performance while using less than 3.8% of trainable parameters, validating both the effectiveness and efficiency of FocA.

II. Related Work

A. RS Image–Text Retrieval

1) Traditional Deep Learning Methods: Early studies primarily adopted the CNN-RNN architecture. For instance, Abdullah et al. [3] proposed TextRS, which utilizes a deep bidirectional triplet network and employs an average fusion strategy to aggregate features from multiple textual descriptions for robust cross-modal matching. However, to address the limitations of global representations, research has shifted

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toward fine-grained alignment mechanisms. Yuan et al. [25] proposed AMFMN, utilizing an adaptive multiscale visual self-attention module to fuse features across different granularities. To reduce spatial semantic ambiguity, Pan et al. [26] developed SWAN, a scene-aware attention network explicitly modeling fine-grained interactions between objects. Similarly, DOVE [8] employs a region-oriented attention mechanism guided by global constraints to ensure accurate alignment between local visual regions and textual words. Although these methods improve performance through architectural enhancements, they mostly rely on single-modal pretraining weights, lacking universal semantic priors derived from large-scale image–text pairs.

2) CLIP-Based Methods: Leveraging CLIP's powerful zero-shot transfer capabilities has become the dominant trend in RSITR. Initial efforts augmented the CLIP framework with specific architectural modules or optimization objectives designed to encourage the model to focus on local details while preserving pretrained knowledge [27]. For instance, Yang et al. [28] proposed IERR, which constructs an implicit-explicit relation reasoning framework by introducing masked image modeling and masked language modeling, utilizing a multimodal interaction encoder to excavate potential correlations between local tokens. Following this thought, Chen et al. [29] designed CLGSA, incorporating a local key feature alignment module alongside a cross-sample global semantic alignment module to further reinforce the unity of local and global semantics. This method enhances the consistent perception of fine-grained semantics via masked reconstruction tasks and hard sample triplet losses. These refined modular designs effectively alleviate the semantic granularity mismatch.

3) Fully Fine-Tuning on Large-Scale RS Data: To fundamentally bridge the domain gap between natural and RS images, another research direction has shifted toward pretraining on extensive RS datasets. Zhang et al. [13] pioneered GeoRSCLIP by constructing the RS5M dataset, which contains five million samples generated via automatic text generation methods, and conducted full-parameter fine-tuning on CLIP to impart domain knowledge previously absent. Subsequently, Liu et al. [30] proposed RemoteCLIP, further integrating multisource data to significantly enhance foundational model capabilities through large-scale data scaling. However, such approaches often entail high computational costs and parameter redundancy.

B. Parameter-Efficient Fine-Tuning

PEFT aims to efficiently adapt pretrained models to downstream tasks by freezing the backbone and updating only a few parameters. Existing PEFT methods can be broadly categorized into three paradigms: adapter-based methods that insert lightweight modules into Transformer layers, LoRA-based methods [15] that inject trainable low-rank matrices into frozen attention weights, and prompt tuning methods [16] that prepend learnable tokens to the input or each Transformer layer. In the context of RSITR, Wang et al. [31] proposed CUP, transferring CLIP knowledge through RS-specific prompt tokens designed to better capture object-level cues. Yang et al. [32] proposed PERT-RaMa, combining low-rank decomposition with Kronecker-product reparameterization to adapt CLIP to RSITR efficiently, incurring low parameter overhead. Nevertheless, both LoRA and prompt tuning adapt the frozen backbone indirectly and do not introduce dedicated trainable modules with internal architectures tailored to specific representational requirements. On the contrary, adapter-based designs insert independent trainable submodules within the backbone, whose structure can be purposefully customized to enhance fine-grained feature extraction and cross-modal alignment, both of which prove critical for capturing fine-grained semantics in RS imagery. Therefore, we focus our subsequent discussion on adapter-based designs, which have emerged as the focal point of recent PEFT research in RSITR. In the general domain, Houlsby et al. [14] introduced the adapter architecture, injecting new knowledge by inserting lightweight bottleneck modules into intermediate Transformer layers. Building on this, AdaptFormer [33] improves efficiency by introducing a parallel bottleneck branch alongside the MLP block; however, its tokenwise independent processing limits the ability to model contextual relationships among patches. Cross-modal adapter [23] uses partially shared up-projection weights for implicit cross-modal alignment. A key limitation is that it conducts alignment without a dedicated feature-enhancement stage, leaving fine-grained details across the two modalities insufficiently refined and aligned. UniAdapter [24] adopts a unified adapter structure for jointly adapting visual and textual encoders across different pretraining frameworks, but its design targets generic cross-modal transfer rather than fine-grained feature extraction.

In the RS domain, Yuan et al. [34] proposed PE-RSITR, which designs a multimodal RS adapter with shared projection to enable effective cross-modal transfer. Nevertheless, its adaptation modules rely on linear projections and lack explicit modeling of contextual relationships among patches. Huang [17] further proposed HarMA, a gating-based hierarchical adapter mechanism that adaptively balances pretrained features with domain-specific features; however, its alignment objective is computed based on the final image and text embeddings and lacks explicit supervision for intermediate representations, causing fine-grained cues to be easily diluted. More recently, Li et al. [35] proposed FGVLA, which integrates fine-grained alignment strategies with spatial mask losses, achieving SOTA performance; however, its alignment depends on noun extraction by external tools, rendering the overall pipeline non-end-to-end and tying alignment quality to the accuracy of noun extraction.

On the contrary to the above methods, FocA emphasizes the capture, alignment, and preservation of fine-grained semantics at intermediate network layers, which is particularly advantageous for fine-grained RS understanding that requires modeling attribute-level and relational-level details.

III. Method

The proposed FocA is designed to bridge the granularity gap between general natural imagery and complex RS scenes. As illustrated in Fig. 2, the framework comprises three core parts: image–text feature extraction using the CLIP backbone, fine-grained feature enhancement and implicit alignment through

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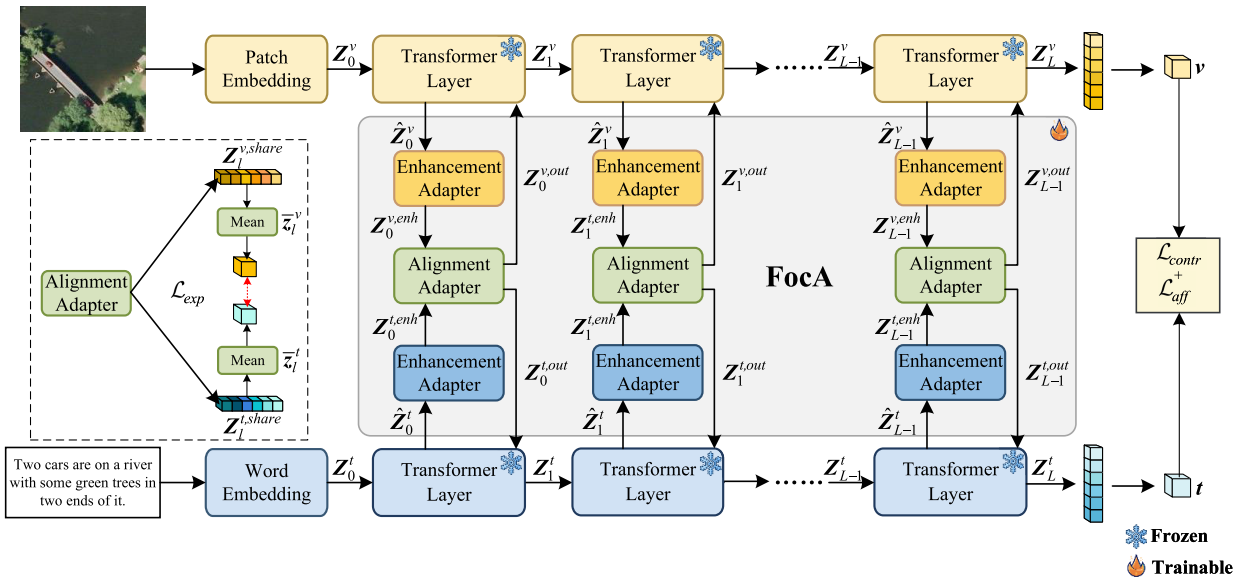


Fig. 2. Overall architecture of the proposed FocA framework. Built upon the frozen CLIP backbone, FocA modules are inserted between the MSA and MLP sublayers across all Transformer layers to enhance fine-grained feature perception and facilitate implicit cross-modal alignment. In addition, an explicit shared loss ($\mathcal{L}_{exp}$) is applied to specific salient intermediate layers to provide direct semantic supervision and enforce explicit alignment.

training FocA, and feature-explicit alignment guided by the loss functions. We discuss each part below.

A. Image–Text Feature Extraction

Given an image–text pair to be aligned, we employ the frozen CLIP backbone's image and text encoders to extract visual and textual features, respectively. Within each Transformer layer of the backbone, FocA enhances the feature extraction process to capture critical fine-grained details. Finally, the extracted visual and textual features are aligned based on the loss functions.

1) Image Encoder: We utilize a vision Transformer [36] as the visual encoder. Given an RS image $\boldsymbol{I} \in \mathbb{R}^{H\times W\times C}$, where H, W, C represent the height, width, and number of channels, respectively. The image is reshaped into a sequence of N flattened 2-D patches. These patches are then linearly projected to an embedding dimension D_v. Next, a learnable class token is prepended to this sequence, and spatial position embeddings are added to form the final input sequence $\boldsymbol{Z}_0^v$.

The sequence $\boldsymbol{Z}_0^v$ is then processed by L stacked Transformer layers. The computation of the lth visual layer can be expressed as

$$\hat{\boldsymbol{Z}}_{l-1}^{v} = \mathrm{MSA}\left(\mathrm{LN}\left(\boldsymbol{Z}_{l-1}^{v}\right)\right) + \boldsymbol{Z}_{l-1}^{v} \tag{1}$$

$$\boldsymbol{Z}_{l-1}^{v,out} = \mathrm{FocA}\left(\hat{\boldsymbol{Z}}_{l-1}^{v}\right) \tag{2}$$

$$\boldsymbol{Z}_{l}^{v} = \mathrm{MLP}\left(\mathrm{LN}\left(\boldsymbol{Z}_{l-1}^{v,out}\right)\right) + \boldsymbol{Z}_{l-1}^{v,out} \tag{3}$$

where MSA(·), MLP(·), LN(·), and FocA(·) denote multihead self-attention, multilayer perceptron, layer normalization, and FocA, respectively.

2) Text Encoder: In parallel with the visual branch, the text encoder also employs a Transformer-based architecture to process natural-language descriptions. Given an input text T, it is first tokenized into a sequence of tokens. Special tokens [SOS] and [EOS] are then appended to the beginning and end of the sequence, respectively. Combined with position embeddings, this yields the final input embedding sequence $\boldsymbol{Z}_0^t$.

Similar to the image encoder, the text features $\boldsymbol{Z}_0^t$ are then processed by L Transformer layers

$$\hat{\boldsymbol{Z}}_{l-1}^{t} = \mathrm{MSA}\left(\mathrm{LN}\left(\boldsymbol{Z}_{l-1}^{t}\right)\right) + \boldsymbol{Z}_{l-1}^{t} \tag{4}$$

$$\boldsymbol{Z}_{l-1}^{t,out} = \mathrm{FocA}\left(\hat{\boldsymbol{Z}}_{l-1}^{t}\right) \tag{5}$$

$$\boldsymbol{Z}_{l}^{t} = \mathrm{MLP}\left(\mathrm{LN}\left(\boldsymbol{Z}_{l-1}^{t,out}\right)\right) + \boldsymbol{Z}_{l-1}^{t,out}. \tag{6}$$

B. Focused Adapter

FocA is the core engine of our framework. It is inserted between the MSA and MLP sublayers within the frozen Transformer blocks of the backbone to transfer general domain knowledge to the RS domain while explicitly focusing on fine-grained attributes inherent in RS imagery. The detailed structure of FocA is shown in Fig. 3. It employs a hybrid architecture consisting of an Enhancement Adapter and an Alignment Adapter. We discuss each adapter below.

1) Enhancement Adapter: $\hat{\boldsymbol{Z}}_{l-1}^{v}$ and $\hat{\boldsymbol{Z}}_{l-1}^{t}$ in (2) and (5) denote the input visual and text features to the Enhancement Adapter in the lth Transformer layer. Both $\hat{\boldsymbol{Z}}_{l-1}^{v}$ and $\hat{\boldsymbol{Z}}_{l-1}^{t}$ are first projected into a bottleneck space, then processed through consecutive MSA operations, and finally adaptively fused with the original features via a learnable interpolation gating mechanism.

Since the processing structure for $\hat{\boldsymbol{Z}}_{l-1}^{v}$ and $\hat{\boldsymbol{Z}}_{l-1}^{t}$ is identical within the enhancement adapter, we use $\hat{\boldsymbol{Z}}_{l-1}^{in} \in \mathbb{R}^{N\times D}$ to represent either modality to detail the processing procedure. First, $\hat{\boldsymbol{Z}}_{l-1}^{in}$ is projected into a bottleneck space using a modality-specific down-projection matrix $\boldsymbol{W}_{down} \in \mathbb{R}^{D\times d}$, where d denotes the bottleneck dimension, followed by a GELU activation function to introduce nonlinearity, yielding

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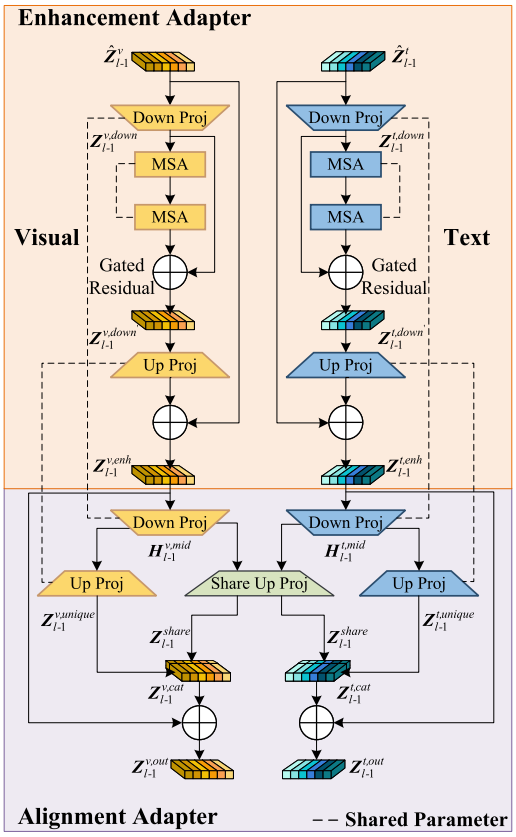


Fig. 3. Detailed structure of the proposed FocA module. It consists of an enhancement adapter for intramodal fine-grained feature mining and an alignment adapter for cross-modal interaction.

the intermediate feature $\boldsymbol{Z}_{l-1}^{down}$. To enhance the feature representation, $\boldsymbol{Z}_{l-1}^{down}$ is fed into two MSA layers with shared parameters. Then, a learnable interpolation gating mechanism is introduced to fuse the enhanced features with $\boldsymbol{Z}_{l-1}^{down}$. Finally, the fused features are projected back to the original dimension D via a modality-specific up-projection matrix $\boldsymbol{W}_{up} \in \mathbb{R}^{d\times D}$ and added to the original input $\hat{\boldsymbol{Z}}_{l-1}^{in}$ through a residual connection, resulting in the enhanced feature $\boldsymbol{Z}_{l-1}^{enh}$. The overall feature enhancement process in the enhancement adapter at the lth Transformer layer can be formulated as

$$\boldsymbol{Z}_{l-1}^{down} = \text{GELU}\left(\hat{\boldsymbol{Z}}_{l-1}^{in}\boldsymbol{W}_{down}\right) \tag{7}$$

$$\alpha = sigmoid\,(\gamma) \tag{8}$$

$$\boldsymbol{Z}_{l-1}^{down'} = (1-\alpha)\cdot \text{MSA}^2\left(\boldsymbol{Z}_{l-1}^{down}\right) + \alpha\cdot \boldsymbol{Z}_{l-1}^{down} \tag{9}$$

$$\boldsymbol{Z}_{l-1}^{enh} = \boldsymbol{Z}_{l-1}^{down'}\boldsymbol{W}_{up} + \hat{\boldsymbol{Z}}_{l-1}^{in} \tag{10}$$

where $\text{MSA}^2(\cdot)$ denotes the operation of two MSA layers with shared parameters applied sequentially. Here, $\gamma \in \mathbb{R}$ is a learnable gating parameter shared across all tokens, and $\alpha \in (0,1)$ is its activated counterpart that controls the interpolation ratio between the MSA-enhanced features and the original bottleneck features.

2) Alignment Adapter: The alignment adapter incorporates a cross-modal shared projection to enforce implicit alignment. To improve parameter efficiency, this stage reuses the projection matrices $\boldsymbol{W}_{down}$ and $\boldsymbol{W}_{up}$ for mapping enhanced features back into the bottleneck space. The enhanced feature $\boldsymbol{Z}_{l-1}^{enh}$ is first projected to $\boldsymbol{H}_{l-1}^{mid}$ via $\boldsymbol{W}_{down}$. Then, the information flow of $\boldsymbol{H}_{l-1}^{mid}$ splits into two paths: one aims to restore modality-specific features, while the other employs a shared linear layer to extract common semantic patterns across modalities.

In the first path, $\boldsymbol{H}_{l-1}^{mid}$ is projected back to the original dimension using the reused $\boldsymbol{W}_{up}$ to generate the modality-unique feature $\boldsymbol{Z}_{l-1}^{unique} \in \mathbb{R}^{N\times D}$. In the second path, $\boldsymbol{H}_{l-1}^{mid}$ is processed by a shared linear layer parameterized by $\boldsymbol{W}_{share} \in \mathbb{R}^{d\times d_s}$, where d_s denotes the dimension of the shared subspace, to extract common semantic patterns, producing the aligned shared feature $\boldsymbol{Z}_{l-1}^{share}$. Notably, to implicitly align the visual and textual representations, a feature splicing strategy is employed: the final segment of the modality-unique feature is replaced with the shared aligned feature to form the implicit aligned feature $\boldsymbol{Z}_{l-1}^{cat}$. This enforces that a designated subspace of the final representation is shared across modalities, while preserving modality-unique characteristics in the remaining dimensions.

Finally, to preserve original information, we employ a residual connection by adding the implicit aligned feature $\boldsymbol{Z}_{l-1}^{cat}$ to the enhanced feature $\boldsymbol{Z}_{l-1}^{enh}$. The computation of the alignment adapter at the lth Transformer layer can be expressed as

$$\boldsymbol{H}_{l-1}^{mid} = \boldsymbol{Z}_{l-1}^{enh}\boldsymbol{W}_{down} \tag{11}$$

$$\boldsymbol{Z}_{l-1}^{unique} = \boldsymbol{H}_{l-1}^{mid}\boldsymbol{W}_{up} \tag{12}$$

$$\boldsymbol{Z}_{l-1}^{share} = \boldsymbol{H}_{l-1}^{mid}\boldsymbol{W}_{share} \tag{13}$$

$$\boldsymbol{Z}_{l-1}^{cat} = \text{Concat}\left(\boldsymbol{Z}_{l-1}^{unique}[1:D-d_s],\ \boldsymbol{Z}_{l-1}^{share}\right) \tag{14}$$

$$\boldsymbol{Z}_{l-1}^{out} = \boldsymbol{Z}_{l-1}^{cat} + \boldsymbol{Z}_{l-1}^{enh} \tag{15}$$

where $\text{Concat}(\cdot,\cdot)$ denotes the concatenation operation along the channel dimension. For the visual and text branches, the output $\boldsymbol{Z}_{l-1}^{out}$ is, respectively, denoted as $\boldsymbol{Z}_{l-1}^{v,out}$ and $\boldsymbol{Z}_{l-1}^{t,out}$.

Finally, the [CLS] and [EOS] tokens from the last layer's outputs are extracted and projected into a shared embedding space. These projected tokens are then L_2-normalized to construct the final image and text features $\boldsymbol{v}$ and $\boldsymbol{t}$.

C. Loss Function

To ensure that the final extracted representations are both discriminative and well-aligned, we design a composite training objective. It comprises three components: a global contrastive loss $\mathcal{L}_{contr}$ for high-level matching, an affiliation loss $\mathcal{L}_{aff}$ to maintain class-specific structure, and the proposed explicit shared loss $\mathcal{L}_{exp}$ to impose direct semantic supervision.

The total loss is defined as

$$\mathcal{L}_{total} = \mathcal{L}_{contr} + \mathcal{L}_{aff} + \mathcal{L}_{exp}. \tag{16}$$

1) Global Contrastive Loss: Following standard VLMs paradigms, we employ the symmetric InfoNCE loss [37] to align the final features of image–text pairs. Given a batch of B image–text pairs, $\boldsymbol{v}_i$ and $\boldsymbol{t}_i$ denote the final visual and text features extracted by the proposed framework from the ith image–text pair. The image-to-text contrastive loss is defined as

$$\mathcal{L}_{contr}^{i2t} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp\left(\boldsymbol{v}_i^{\top}\boldsymbol{t}_i/\tau\right)}{\sum_{j=1}^{B}\exp\left(\boldsymbol{v}_i^{\top}\boldsymbol{t}_j/\tau\right)} \tag{17}$$

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where τ is a learnable temperature parameter. Meanwhile, the text-to-image loss $\mathcal{L}_{contr}^{t2i}$ is computed symmetrically, and the total global contrastive loss is $\mathcal{L}_{contr} = (\mathcal{L}_{contr}^{i2t} + \mathcal{L}_{contr}^{t2i})/2$.

2) Affiliation Loss: To mitigate the potential increase in intraclass distance, we adopt the affiliation loss [38]. By aligning samples to their corresponding category centers, this loss encourages the model to preserve a structured and compact global feature space, even as it attends to distinct fine-grained attributes. The image-to-text affiliation loss is defined as

$$\mathcal{L}_{aff}^{i2t} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp\left(\boldsymbol{v}_i^\top \boldsymbol{t}_i^*/\tau\right)}{\sum_{j=1}^{B}\exp\left(\boldsymbol{v}_i^\top \boldsymbol{t}_j^*/\tau\right)} \tag{18}$$

where $\boldsymbol{t}_i^*$ represents the cluster center corresponding to the category of the ith text. The text-to-image affiliation loss $\mathcal{L}_{aff}^{t2i}$ is computed symmetrically by aligning text features to their corresponding visual cluster centers. The total affiliation loss is $\mathcal{L}_{aff} = (\mathcal{L}_{aff}^{i2t} + \mathcal{L}_{aff}^{t2i})/2$.

3) Explicit Shared Loss: To impose direct semantic supervision on intermediate representations, we design an explicit shared loss. This loss imposes direct supervision on specific intermediate layers, explicitly aligning modalities at the precise stage where fine-grained semantics are most salient. For the lth layer where the explicit shared loss is applied, we extract the enhanced visual and textual tokens output by the corresponding alignment adapter. Then, we perform global average pooling on the output tokens to obtain the layer-specific holistic representations $\bar{\boldsymbol{z}}_l^v$ and $\bar{\boldsymbol{z}}_l^t$

$$\bar{\boldsymbol{z}}_l^v = \text{Mean}\left(\boldsymbol{Z}_l^{v,share}\right),\ \bar{\boldsymbol{z}}_l^t = \text{Mean}\left(\boldsymbol{Z}_l^{t,share}\right). \tag{19}$$

We then apply the symmetric contrastive loss to these intermediate vectors. The explicit shared loss is defined as

$$\mathcal{L}_{exp} = \sum_{l\in L}\frac{1}{2}\left(\mathcal{L}^{i2t}\left(\bar{\boldsymbol{z}}_l^v, \bar{\boldsymbol{z}}_l^t\right) + \mathcal{L}^{t2i}\left(\bar{\boldsymbol{z}}_l^t, \bar{\boldsymbol{z}}_l^v\right)\right). \tag{20}$$

IV. Experiment

A. Datasets and Evaluation Metrics

To evaluate the effectiveness of the proposed method, we conduct experiments on two widely used datasets: RSICD [39] and RSITMD [25]. RSICD contains 10 921 images and 54 605 captions, characterized by high intraclass diversity, which serves to validate the model's global semantic understanding. RSITMD comprises 4743 images and 23 715 fine-grained descriptions, utilized to verify the sensitivity to local details and object-level alignment. Following previous methods [17], we adopt the standard partition for both datasets. In particular, RSICD is divided into 7862 training, 1966 validation, and 1093 testing images, while RSITMD is split into 3435 training, 856 validation, and 452 testing images.

For quantitative evaluation, we adopt recall at K ($R@K$, with $K = 1, 5, 10$) and mean recall (mR) as metrics. $R@K$ measures the percentage of queries where the ground truth is successfully retrieved within the top-K results. The mR is computed by averaging all $R@K$ scores across both image-to-text and text-to-image retrieval tasks, providing an overall assessment of the model's performance.

We further evaluate the RSITR methods on the semantic localization (SeLo) task [40], [41], which uses text queries to locate semantically relevant regions in large-scale RS scenes, to verify their effectiveness in aligning RS images with text. The RSITR methods are fine-tuned on the RSITMD dataset and then tested on the AIR-SLT test set. According to the official settings, SeLo v1 performs sliding window cropping at three scales of 256 × 256, 512 × 512, and 768 × 768, whereas SeLo v2 first performs single-scale 512 × 512 coarse cropping and then applies multipoint multiscale expansion using 128 × 128 and 256 × 256 crops around high-likelihood regions. Four SeLo metrics are employed: significant area proportion (R_{su}), attention shift distance (R_{as}), discrete attention distance (R_{da}), and the mean indicator (R_{mi}). In particular, R_{su} measures the ratio of attention within the ground-truth (GT) regions over the non-GT regions; R_{as} quantifies the offset of the attention focus relative to the GT center; R_{da} evaluates the discreteness of the generated attention based on probability divergence and candidate count; and R_{mi} is a weighted combination of the three. All metrics range from 0 to 1, with higher values indicating better performance, except for R_{as}.

B. Implementation Details

In this work, the proposed framework is built upon the pretrained CLIP ViT-B/32 model. All training and inference processes are conducted on a server equipped with 4 NVIDIA GeForce RTX 3090 GPUs. We employ the AdamW optimizer with an initial learning rate of 4×10^{-4} and a weight decay of 0.04. The global batch size is set to 214. The training duration is set to 6 epochs for RSICD and 10 epochs for RSITMD. Regarding the explicit shared loss $\mathcal{L}_{exp}$, it is applied at the second layer for the RSICD dataset and the eighth layer for the RSITMD dataset. In addition, the initial value for the temperature parameter τ is set to 0.07, the learnable gating parameter γ in the adapter module is initialized to 0.6, and the bottleneck dimension d and the shared subspace dimension d_s are both set to 128. The total trainable parameters amount to 0.48 M, with the enhancement adapter accounting for approximately 0.46 M and the alignment adapter about 0.02 M. Notably, the alignment adapter remains lightweight by reusing projection matrices from the enhancement adapter.

C. Comparison With SOTA Methods

We evaluate FocA against 16 image–text retrieval methods on the RSICD and RSITMD datasets. As summarized in Table I, these methods are categorized into two groups: traditional methods and CLIP-based methods. Across both datasets, our method consistently outperforms most methods in terms of both $R@K$ and mR metrics, demonstrating the superiority of our focused adaptation strategy.

1) Traditional Methods: Feature extractors used in traditional methods are typically based on deep networks with nontransformer architectures, such as CNNs and RNNs. VSE++ [42] introduces a hard negative mining strategy to emphasize difficult samples. AMFMN [25] employs a multiscale visual self-attention module to fuse features at different

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TABLE I

PERFORMANCE COMPARISON WITH SOTA METHODS ON RSICD AND RSITMD DATASETS. BOLD INDICATES THE BEST PERFORMANCE AND UNDERLINED VALUES INDICATE THE SECOND BEST

Methods	Ref.	Params	RSICD							RSITMD						
			Image-to-Text			Text-to-Image			mR	Image-to-Text			Text-to-Image			mR
			R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
Traditional methods																
VSE++ [42]	BMVC'18	15.78M	3.38	9.51	17.46	2.82	11.32	18.10	10.43	10.38	27.65	39.60	7.79	24.87	38.87	24.83
AMFMN [25]	TGRS'22	13.68M	4.39	10.90	17.64	3.91	16.20	26.49	13.25	10.63	24.78	41.81	11.51	34.09	54.87	29.72
HVSA [43]	TGRS'23	-	7.47	20.62	32.11	5.51	21.13	34.13	20.16	13.20	32.08	45.58	11.43	39.20	57.45	33.16
SWAN [26]	ICMR'23	40M	7.41	20.13	30.86	5.56	22.26	37.41	20.61	13.35	32.15	46.90	11.24	40.40	60.60	34.11
DOVE [8]	TGRS'24	38M	8.66	22.35	34.95	6.04	23.95	40.35	22.72	16.81	36.80	49.93	12.20	44.13	66.50	37.73
CLIP-based Methods																
Adapter [14]	ICML'19	0.17M	8.73	24.73	37.81	8.43	26.02	43.33	24.84	13.75	27.64	39.96	12.89	40.09	59.91	32.37
CLIP-Adapter [44]	IJCV'24	0.52M	7.11	19.48	31.01	7.67	24.87	39.73	21.65	12.83	28.84	39.05	13.30	40.20	60.06	32.38
AdaptFormer [33]	NIPS'22	0.17M	12.46	28.49	41.86	9.09	29.89	46.81	28.10	16.71	30.16	42.91	14.27	41.53	61.46	34.81
Cross-Modal Adapter [23]	PR'25	0.16M	11.18	27.31	40.62	9.57	30.74	48.36	27.96	18.16	36.08	48.72	16.31	44.33	64.75	38.06
UniAdapter [24]	ICLR'24	0.55M	12.65	30.81	42.74	9.61	30.06	47.16	28.84	19.86	36.32	51.28	17.54	44.89	56.46	39.23
R-Adapter [45]	ECCV'24	1.70M	10.45	32.48	48.20	14.00	31.29	43.64	30.01	16.55	46.81	**64.69**	21.24	42.70	53.98	41.00
PERT-RaMa [32]	TGRS'25	0.53M	12.97	31.53	45.20	11.47	34.04	51.46	31.11	20.66	44.16	58.94	17.95	50.81	69.08	43.60
PE-RSITR [34]	TGRS'23	0.16M	14.13	31.51	44.78	11.63	33.92	50.73	31.12	23.67	44.07	60.36	20.10	50.63	67.97	44.47
HarMA [17]	ICLRW'24	0.50M	16.36	34.48	47.74	12.92	37.17	53.07	33.62	25.81	48.37	60.61	19.92	53.27	71.21	46.53
CLGSA [29]	TGRS'25	-	14.38	36.73	45.35	11.75	34.07	50.12	32.07	26.77	48.23	60.39	22.52	52.39	69.46	46.63
FGVLA [35]	TGRS'25	0.46M	17.20	36.41	49.95	12.77	36.78	53.60	34.45	25.22	49.34	64.16	21.73	54.51	**72.48**	47.91
Ours	-	0.48M	**20.49**	**41.35**	**53.43**	**14.49**	**39.23**	**56.63**	**37.61**	**27.21**	**51.11**	63.50	**22.88**	**56.55**	**72.48**	**48.95**

granularities. HVSA [43] learns hypersphere-based visual-semantic alignment to capture correspondences at multiple levels. SWAN [26] focuses on reducing spatial semantic confusion by establishing scene-aware correlations across features. DOVE [8] utilizes a region-oriented attention mechanism guided by global semantic constraints to enhance alignment.

2) CLIP-Based Methods: These methods leverage pretrained knowledge from CLIP and transfer it to the RS domain via fine-tuning or adapter strategies. Adapter [14] inserts lightweight bottleneck layers sequentially into the frozen backbone to adapt representations. CLIP-Adapter [44] utilizes a residual connection with two linear layers for feature adaptation. AdaptFormer [33] employs a parallel branch design to efficiently update the MLP blocks. Cross-modal adapter [23] facilitates implicit cross-modal interactions at the encoder level. UniAdapter [24] inserts unified layers into the backbone for pretraining-agnostic adaptation. R-Adapter [45] incorporates a self-ensemble technique with lightweight modules to boost robustness. PERT-RaMa [32] employs parameter-efficient reparameterization tuning with a ranking and matching training strategy. PE-RSITR [34] designs modality-specific adapters for parameter-efficient transfer learning. HarMA [17] introduces a hierarchical mini-adapter with a gating mechanism to balance pretrained and domain-specific features. CLGSA [29] explores context-aware local-global semantic alignment to improve retrieval accuracy. FGVLA [35] employs a fine-grained visual–language alignment strategy with spatial mask losses.

3) Results on RSICD Dataset: The comparison results on the RSICD dataset are presented in the left half of Table I. Clearly, CLIP-based methods demonstrate a significant advantage over traditional deep learning architectures in both retrieval accuracy and parameter efficiency. Traditional methods require a substantial number of parameters, ranging from 13.68 to 40 M, yet only achieve a maximum *mR* of 22.72%. On the contrary, CLIP-based methods leverage pretrained vision–language priors to achieve higher accuracy with significantly fewer trainable parameters, often utilizing less than 1 M. For example, the FGVLA achieves an *mR* of 34.45% while utilizing only 0.46-M parameters, underscoring the efficiency of the visual–language pretraining paradigm.

The proposed FocA method achieves the highest performance on RSICD with an *mR* of 37.61%, surpassing the SOTA FGVLA by 3.16%. In particular, our method improves text retrieval *R*@1 by +3.29% and image retrieval R@1 by +1.72% compared to FGVLA. Notably, FocA maintains a very low parameter count of only 0.48 M. These improvements are attributed to the unique hybrid adapter architecture and the introduction of an explicit shared loss.

4) Results on RSITMD Dataset: A similar trend is observed on the fine-grained RSITMD dataset, where CLIP-based methods consistently outperform traditional models. Traditional architectures peak at an *mR* of 37.73%, while CLIP-based approaches like HarMA and FGVLA reach 46.53% and 47.91%, respectively. This performance gap highlights the difficulty traditional models face when attempting to capture the subtle object-level details and complex backgrounds inherent in scene-centric RS imagery.

Our method establishes a new SOTA on RSITMD with a record *mR* of 48.95%. It achieves notable gains of +1.99%

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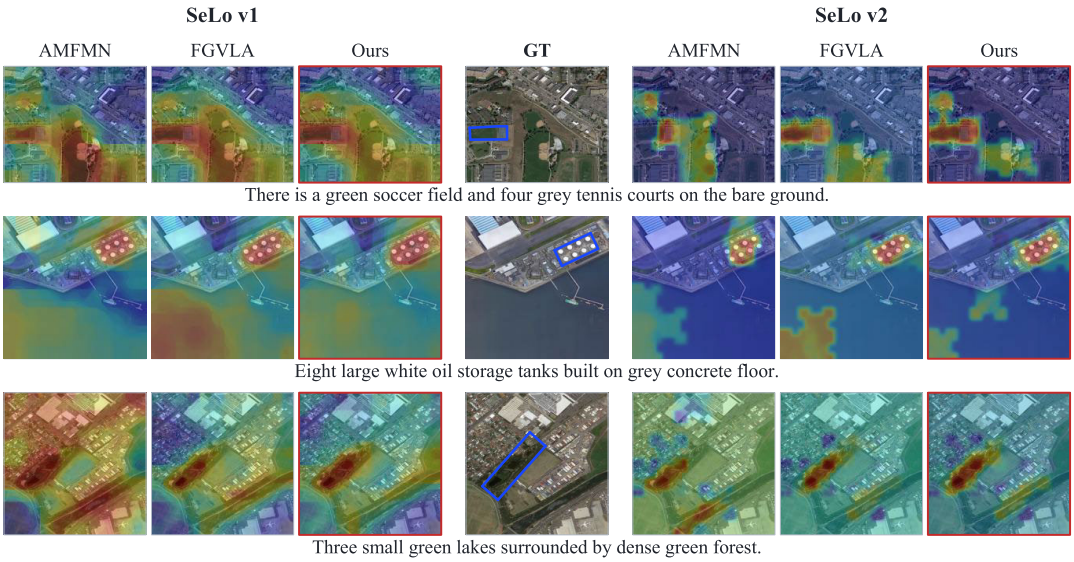


Fig. 4. Comparison of visualizations from AMFMN, FGVLA, and our proposed FocA on the SeLo v1 and v2 benchmarks.

TABLE II
PERFORMANCE COMPARISON ON THE SELO V1 BENCHMARK

Methods	$R_{su}\uparrow$	$R_{da}\uparrow$	$R_{as}\downarrow$	$R_{mi}\uparrow$
AMFMN [25]	0.692	0.667	0.332	0.677
FGVLA [35]	0.759	0.709	0.280	0.733
FocA (Ours)	**0.767**	**0.717**	**0.266**	**0.743**

TABLE III
PERFORMANCE COMPARISON ON THE SELO V2 BENCHMARK

Methods	$R_{su}\uparrow$	$R_{da}\uparrow$	$R_{as}\downarrow$	$R_{mi}\uparrow$
AMFMN [25]	0.720	**0.666**	**0.293**	0.702
FGVLA [35]	0.797	0.663	0.299	**0.730**
FocA (Ours)	**0.809**	0.626	0.294	0.727

in text retrieval $R@1$ and +1.15% in image retrieval R@1 over FGVLA. Furthermore, in the image retrieval task, FocA outperforms the strong HarMA competitor by a margin of +2.96% in $R@1$. These results are driven by the Alignment adapter's ability to bridge the modality gap through a shared projection subspace. This mechanism facilitates implicit cross-modal interaction at intermediate layers where fine-grained semantics are most salient, allowing the model to accurately align visual entities with complex textual attributes in highly detailed scenes.

D. Performance on SeLo

We evaluate FocA on the SeLo v1 and v2 benchmarks. For comparison, we select the leading CLIP-based method, FGVLA, and the standard baseline in the SeLo task, AMFMN. Table II summarizes the results on the SeLo v1 benchmark, where FocA outperforms both competing methods across all four evaluation metrics. This demonstrates that under the SeLo v1 benchmark, FocA effectively achieves robust semantic alignment between the text and image modalities.

Regarding the SeLo v2 benchmark, its incorporation of multilevel likelihood expansion (MLE) pruning and the Chaotic self-feeding mechanism (CSM) for feature stabilization accelerates inference while alleviating the background interference caused by the exhaustive traversal employed in SeLo v1. Consequently, all evaluated algorithms observe varying degrees of improvement in R_{su} (see Table III). However, the resulting attention activations become more fine-grained, leading to a more dispersed distribution rather than strict concentration around the geometric center of the ground truth (see Fig. 4). This results in a deterioration of both R_{da} and R_{as} across all methods (except for R_{as} of AMFMN)—an effect that is particularly pronounced in FocA due to its intrinsically finer-grained feature representations.

We further provide a qualitative comparison of localization results produced by AMFMN, FGVLA, and our proposed FocA in Fig. 4. Overall, the activation maps generated under the SeLo v2 benchmark exhibit noticeably finer-grained semantic representations across all methods compared to those from SeLo v1. In particular, in the first example, FocA precisely anchors its attention on the "green soccer field and four gray tennis courts," whereas under SeLo v1, both AMFMN and FGVLA mistakenly assign high activations to the adjacent baseball field. In the second example, FocA's response is tightly concentrated on the "eight large white oil storage tanks," whereas AMFMN's activation does not cover the target region indicated by the text, and FGVLA incorrectly activates the sea area. Similarly, in the third example, FocA achieves substantially sharper and more focused grounding of the "three small green lakes" compared to the other two methods.

These results collectively validate the efficacy of our fine-grained semantic design in FocA for the SeLo task.

E. Analysis of Training Dynamics

To verify that the three loss components ($\mathcal{L}_{contr}$, $\mathcal{L}_{aff}$, and $\mathcal{L}_{exp}$) in our composite training objective optimize compatibly

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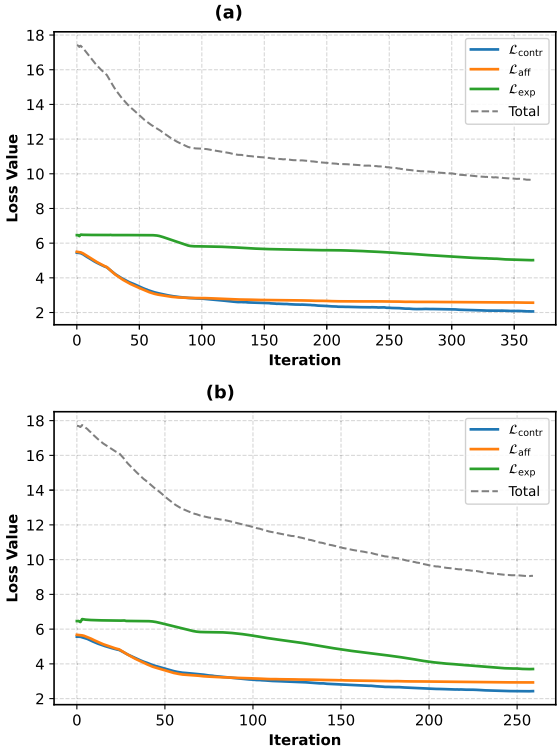


Fig. 5. Training loss curves on (a) RSICD and (b) RSITMD datasets.

without gradient conflicts or inconsistent objectives, we analyze the training loss dynamics on both RSICD and RSITMD datasets, as shown in Fig. 5.

Across both datasets, all three loss components decrease monotonically throughout training, with the total loss converging smoothly and without oscillations. In particular, $\mathcal{L}_{contr}$ and $\mathcal{L}_{aff}$ decrease in close synchrony, indicating that the optimization directions are aligned and compatible. $\mathcal{L}_{exp}$ exhibits a slightly delayed decline, particularly during the early training phase. This behavior aligns with its design as intermediate-layer supervision: during initial iterations, intermediate representations lack sufficient semantic structure for effective cross-modal alignment, resulting in limited optimization signals from $\mathcal{L}_{exp}$. As training progresses and primary objectives shape these intermediate representations, $\mathcal{L}_{exp}$ becomes more effective, contributing to fine-grained refinement. The smooth and concurrent decrease of all three losses across both datasets indicates an absence of gradient conflicts, thereby validating the compatibility of $\mathcal{L}_{contr}$, $\mathcal{L}_{aff}$, and $\mathcal{L}_{exp}$ within a unified training framework.

F. Ablation Studies

In this section, we conduct detailed ablation experiments on both the RSICD and RSITMD datasets to comprehensively analyze the effectiveness of each component and the impact of key hyperparameters within the proposed FocA architecture. By evaluating the model under various configurations, we aim to validate the contributions of the core modules, verify the importance of each loss function, determine the optimal dimensionality for the shared projection subspace, and identify the most effective layers for applying intermediate supervision.

1) Analysis of Component Contributions: We evaluate the individual and combined contributions of the two core components: the enhancement adapter and the alignment adapter, with quantitative results summarized in Table IV. When evaluating the base configuration containing only the enhancement adapter, the model utilizes 0.46-M parameters and establishes a baseline performance with an *mR* of 36.30% on RSICD and 47.15% on RSITMD. The Alignment Adapter is evaluated in conjunction with the explicit shared loss ($\mathcal{L}_{exp}$) on the frozen backbone, as these two modules are engineered to work in tandem. Under this configuration, the model requires only 0.35-M parameters and achieves an *mR* of 36.76% on RSICD and 48.12% on RSITMD. The performance gains over the configuration using only the enhancement adapter demonstrate the effectiveness of this approach in bridging the modality gap at intermediate layers, where visual and textual representations preserve their most salient fine-grained semantics.

The combination of both the enhancement and alignment adapters, with a parameter count of only 0.48 M, yields consistent improvements across both datasets. The *mR* scores increase to 37.61% on RSICD and 48.95% on RSITMD. This improvement validates the module's effectiveness in bridging the modality gap through a cross-modal shared projection subspace, thereby achieving effective implicit alignment between visual and textual representations.

2) Analysis of Losses: To further investigate the effectiveness of the proposed composite training objective, we conduct ablation studies on the individual loss components: the contrastive loss ($\mathcal{L}_{contr}$), the affiliation loss ($\mathcal{L}_{aff}$), and the explicit shared loss ($\mathcal{L}_{exp}$). As summarized in Table V, using only $\mathcal{L}_{contr}$ as a baseline yields an *mR* of 36.76% on RSICD and 48.16% on RSITMD. Integrating the $\mathcal{L}_{aff}$ helps mitigate intraclass distance and preserves a structured feature space, which improves the *mR* to 37.31% on RSICD and 48.27% on RSITMD. Similarly, the inclusion of the $\mathcal{L}_{exp}$ independently enhances the *mR* to 37.20% and 48.41%, respectively, validating the benefit of providing direct semantic supervision at the intermediate layer.

The optimal performance is achieved when all three loss components are combined, resulting in an *mR* of 37.61% on RSICD and 48.95% on RSITMD. This comprehensive objective ensures that the learned representations are not only well-aligned globally but also maintain a discriminative, category-specific structure while preventing the dilution of fine-grained semantics during deep propagation.

3) Ablation Study of Shared Token Dimension: We investigate the impact of the shared token dimension (d_s) within the alignment adapter, as this parameter controls the capacity of the cross-modal shared projection subspace. As summarized in Table VI, we vary the dimension from 32 to 256 to identify the optimal balance between representation capacity and parameter efficiency. The experimental results indicate that retrieval performance initially improves as the dimension increases, reaching a peak at $d_s = 128$. At this dimensionality, the architecture achieves its highest *mR* of 37.61% on RSICD and 48.95% on RSITMD. While the parameter count increases slightly compared to the 64-D configuration (from 0.469 to 0.477 M), the notable performance gain justifies the

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TABLE IV
ABLATION STUDIES OF DIFFERENT COMPONENTS ON RSICD AND RSITMD DATASETS

Components		Params (M)	RSICD Dataset							RSITMD Dataset						
Enhancement Adapter	Alignment Adapter		Image-to-Text			Text-to-Image			mR	Image-to-Text			Text-to-Image			mR
			R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
✓		0.46	20.22	38.15	50.14	14.33	38.79	56.16	36.30	25.22	47.35	61.95	21.02	54.87	72.52	47.15
	✓	0.35	20.49	39.07	51.24	14.07	**39.45**	56.21	36.76	**27.65**	48.89	61.06	21.33	**56.77**	**73.01**	48.12
✓	✓	0.48	**20.49**	**41.35**	**53.43**	**14.49**	39.23	**56.63**	**37.61**	27.21	**51.11**	**63.50**	**22.88**	56.55	72.48	**48.95**

TABLE V
ABLATION STUDIES OF LOSSES ON RSICD AND RSITMD DATASETS

Loss			RSICD Dataset							RSITMD Dataset						
$\mathcal{L}_{contr}$	$\mathcal{L}_{aff}$	$\mathcal{L}_{exp}$	Image-to-Text			Text-to-Image			mR	Image-to-Text			Text-to-Image			mR
			R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
✓			19.58	38.98	52.33	14.53	39.12	55.99	36.76	**29.42**	47.79	61.50	**23.10**	56.02	71.11	48.16
✓	✓		**20.77**	40.07	**53.61**	13.78	39.45	56.16	37.31	26.77	49.12	**63.72**	22.48	55.80	71.77	48.27
✓		✓	19.30	39.80	52.61	**15.08**	**39.80**	**56.60**	37.20	28.10	48.45	62.17	22.70	**56.90**	72.12	48.41
✓	✓	✓	20.49	**41.35**	53.43	14.49	39.23	56.63	**37.61**	27.21	**51.11**	63.50	22.88	56.55	**72.48**	**48.95**

TABLE VI
ABLATION STUDY OF SHARED TOKEN DIMENSION

Dimension	Params (M)	mR	
		RSICD	RSITMD
32	0.465	37.53	48.72
64	0.469	37.56	48.75
128	0.477	**37.61**	**48.95**
256	0.494	37.49	48.80

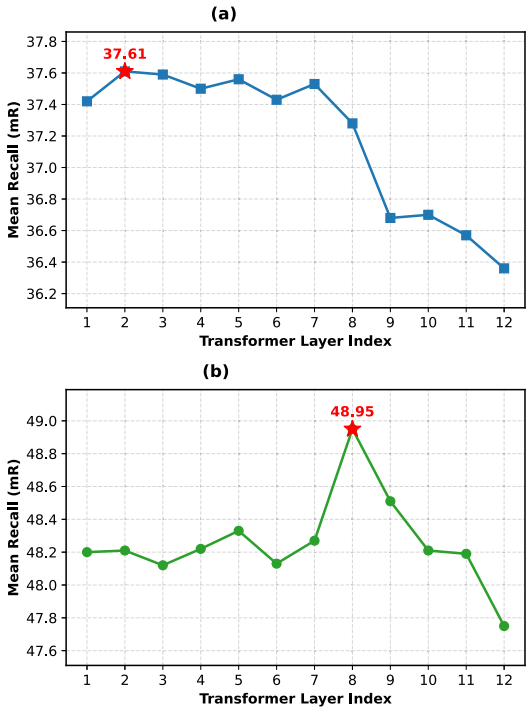


Fig. 6. Parameter sensitivity analysis of the explicit shared loss ($\mathcal{L}_{exp}$) applied at different Transformer layers. The curves show the mR variations on (a) RSICD and (b) RSITMD datasets, highlighting the optimal layers for intermediate supervision.

minor increase in complexity. However, further increasing the shared token dimension to 256 leads to a slight degradation in performance. This decline is likely due to overfitting or the introduction of redundant information within the high-dimensional subspace, which can obscure critical fine-grained features. Based on these observations, we select 128 as the optimal dimension for our final design, as it provides the best tradeoff between model capacity and parameter efficiency.

4) Ablation Study of Layer Selection: We analyze the impact of applying the explicit shared loss at different Transformer layers to identify the optimal stage for intermediate supervision. As illustrated in Fig. 6, the optimal layer index exhibits distinct patterns between the two datasets, reflecting their unique data characteristics. For the RSICD dataset, the best performance is obtained at the second layer with an *mR* of 37.61%, whereas for RSITMD, the optimal performance is achieved at the eighth layer, yielding an *mR* of 48.95%.

We attribute this phenomenon to the intrinsic differences in data complexity and semantic granularity between the two datasets. RSICD typically contains lower-resolution imagery with simpler, scene-level descriptions, which allows the model to extract and align sufficient semantic features in shallower layers. On the contrary, the RSITMD dataset comprises higher-resolution imagery characterized by complex, fine-grained object relations and more detailed textual descriptions. Such complexity requires deeper propagation through the Transformer blocks to fully capture and disentangle the distinctive features necessary for precise cross-modal alignment. Consequently, we apply the explicit shared loss at the second layer

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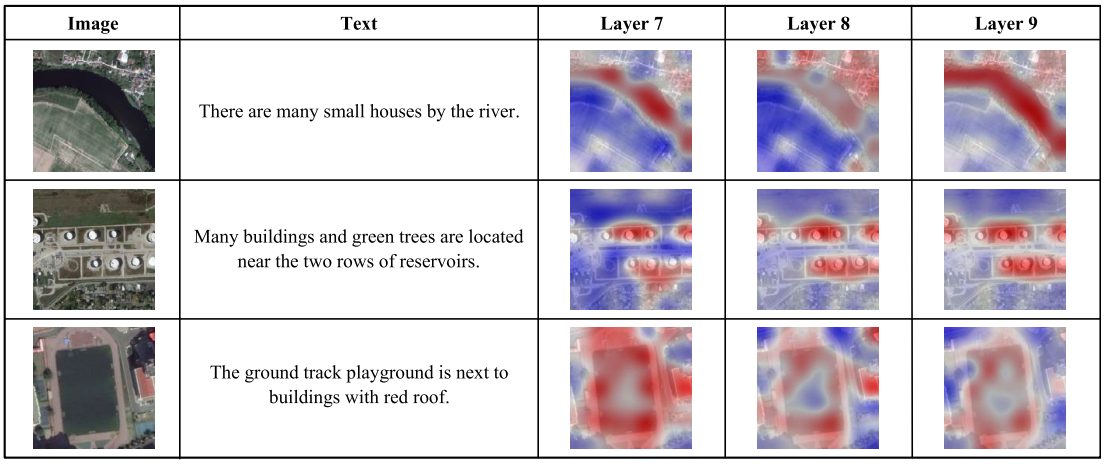


Fig. 7. Visualization of attention difference maps between the FocA and CLIP model at intermediate layers (Layer 7–9). Red regions indicate where FocA exhibits stronger attention than the baseline, while blue regions indicate weaker attention.

for RSICD and the eighth layer for RSITMD to maximize the effectiveness of our intermediate supervision strategy.

G. Qualitative Analysis

1) Visualization of Attention Difference: To intuitively verify the effectiveness of FocA in capturing fine-grained semantics, we visualize the attention difference maps between the proposed method and the CLIP baseline. In particular, we compute the attention maps of the final textual embedding with respect to the visual tokens at intermediate layers 7, 8, and 9, as these layers represent where object-level features are relatively complete and exhibit high distinctiveness. By subtracting the baseline's attention map from ours, we highlight specific improvements; in Fig. 7, red regions indicate where our model exhibits stronger attention than the baseline, while blue indicates weaker activations.

As shown in the first row of Fig. 7, for the query "There are many small houses by the river," our model generates distinct red activations covering both the river and residential areas, proving its superior capability to locate concrete objects. The second row further demonstrates the model's understanding of complex spatial structures; for the description "two rows of reservoirs," the red regions clearly delineate a parallel linear structure, indicating that FocA successfully captures fine-grained spatial and numerical concepts.

Furthermore, the third row highlights our model's sensitivity to specific attributes. Beyond recognizing the "ground track playground," the attention focus explicitly extends to the "buildings with red roofs," demonstrating that the enhancement adapter effectively sharpens the perception of fine-grained color attributes. Collectively, these visualizations confirm that incorporating FocA enables the model to refine the focus of the pretrained backbone, ensuring visual attention is precisely aligned with fine-grained textual details.

2) Visualization of Retrieval Results: To further demonstrate the superiority of FocA in real-world retrieval scenarios, we compare the top-ranked retrieval results of our method against the SOTA FGVLA method. The visualization results, presented in Fig. 8, cover both image-to-text and text-to-image tasks, highlighting the architecture's ability to handle intricate semantic details. In the image-to-text examples, for a scene containing a bridge, our model successfully captures specific visual attributes corresponding to "white iron," enabling it to retrieve all correct GT captions. On the contrary, FGVLA fails to identify these specific textures and retrieves only coarse-grained texts. Similarly, in an example depicting a resort, FocA not only recognizes the scene category but also accurately identifies the "curved" shape of the road, whereas FGVLA fails to match these specific semantics.

In the text-to-image task, when queried with "Three lines of cars are near a line of trees," FocA accurately captures both the concrete objects and the structural description "three lines" to successfully retrieve the target image. Conversely, FGVLA appears to focus primarily on the "trees" while ignoring the critical spatial arrangement of the cars, resulting in incorrect matches. Furthermore, for complex queries describing "three swimming pools" alongside multiple entities such as "buildings," "lawn," and "parking lot," our model exhibits precise multiobject alignment capabilities. It correctly identifies the numerical constraint and the coexistence of multiple entities, whereas FGVLA mainly focuses on the "swimming pool" and neglects the comprehensive context required for accurate matching. These comparisons collectively verify that FocA significantly outperforms the SOTA method in capturing fine-grained details, shapes, and complex spatial relationships.

3) Visualization of Class Activation Maps: To visually interpret the semantic alignment learned by our network, we employ Grad-CAM to visualize the class activation maps for both FocA and the SOTA FGVLA method, as illustrated in Fig. 9. In these visualizations, red regions denote high activation levels, indicating the specific areas where the model focuses its attention given a particular text query.

In the first row, when queried with "blue roofs," FocA accurately highlights the specific blue roofing areas. On the contrary, FGVLA and CLIP fail to distinguish the specific color attributes, incorrectly activating the adjacent gray roofs alongside the blue ones. Similarly, in the second example concerning "large lawns," our model precisely localizes the lawn area. However, CLIP ignores the size constraint and

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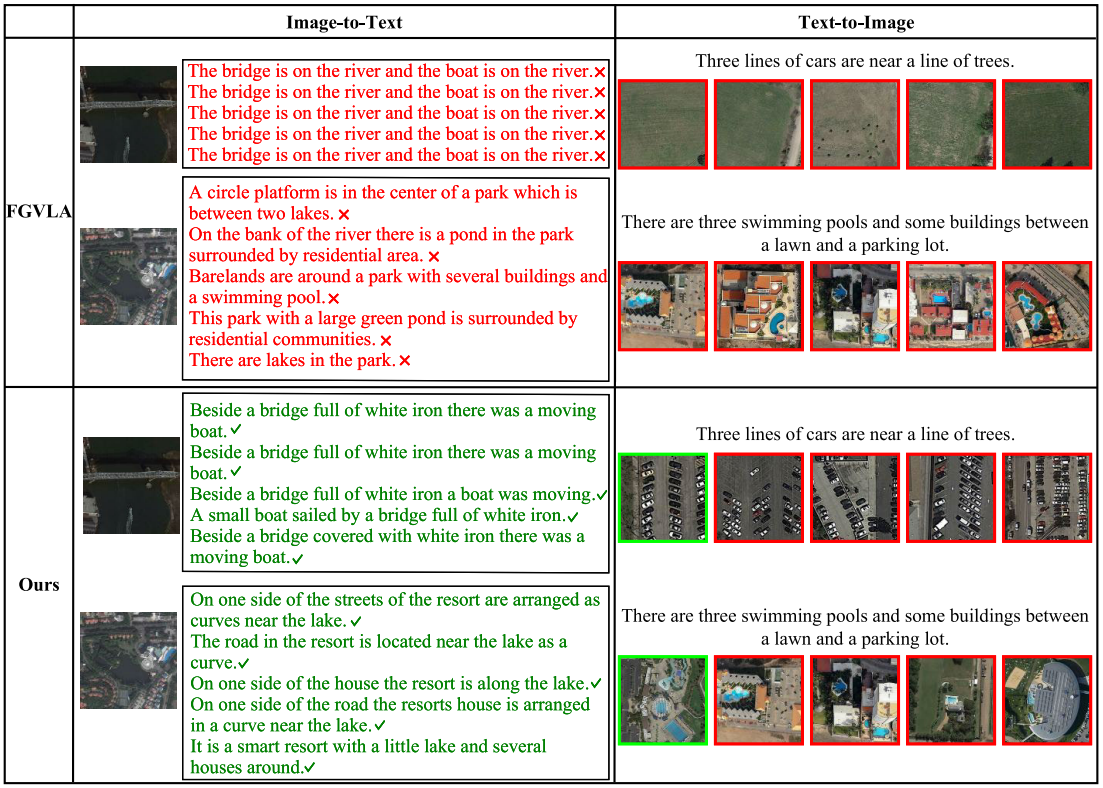


Fig. 8. Qualitative comparison of top-ranked retrieval results between the SOTA method FGVLA and our proposed FocA. Green boxes/ticks indicate correct matches, while red boxes/crosses indicate incorrect ones.

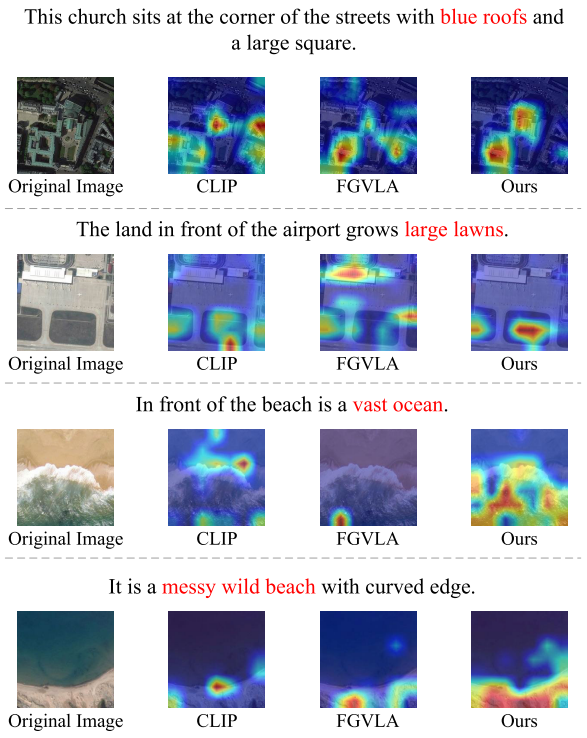


Fig. 9. Visualization of class activation maps (Grad-CAM) comparing the attention focus of CLIP, FGVLA, and FocA.

activates even the small marginal lawns, whereas FGVLA is confused by the adjacent large buildings, failing to discriminate between the object category "lawn" and the size attribute "large."

Most importantly, to evaluate discriminative ability under complex backgrounds, we analyze two distinct queries within the same "beach" scenario. When queried with "vast ocean," our model successfully covers the extensive water surface. However, CLIP incorrectly activates the adjacent beach area, while FGVLA only activates a small, fragmented region. Conversely, when the query shifts to "messy wild beach," our model activates a broad range of the beach area, effectively capturing the extensive scene semantics. In contrast, FGVLA only covers a limited and fragmented region, failing to grasp the holistic semantics. These contrasts strongly demonstrate that FocA not only identifies specific object categories but also effectively disentangles fine-grained attributes, such as color and extent, from complex backgrounds, thereby avoiding the attribute leakage observed in other methods.

V. Conclusion

In this article, we introduced the FocA architecture to address the fine-grained semantic alignment mismatch in RS image–text Retrieval. By integrating modality-specific feature enhancement with a cross-modal shared projection subspace, FocA effectively captures the intricate details and spatial relationships inherent in RS imagery. Our approach mitigates the phenomenon of semantic oversmoothing by applying explicit semantic constraints at intermediate Transformer layers where fine-grained attributes are most salient. Quantitative results on benchmark datasets show that FocA consistently outperforms existing traditional and CLIP-based methods in retrieval

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accuracy and parameter efficiency. Qualitative visualizations further confirm the model's superior ability to disentangle complex attributes like color, shape, and numerical constraints from dense backgrounds.

Nevertheless, the current experiments are only conducted with the CLIP ViT-B/32 backbone on standard RSITR benchmarks. In future work, we plan to extend FocA to larger or RS-specialized backbones (e.g., GeoRSCLIP, RemoteCLIP) and to broader scenarios, such as zero-shot and open-vocabulary retrieval.

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